LESSON 2.9

Describe Data by Mean Absolute Deviation

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the mean absolute deviation (MAD) to describe how spread out data is.

WHY THIS MATTERS The forecast shows the expected high temperatures for a week. The mean high temperature for the week is 79°F. How can you describe the variation in high temperatures over the week?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Mean Absolute Deviation *Spanish: Desviación media absoluta*	The average distance of data values from the mean; it describes spread.	Written as MAD
Deviation *Spanish: Desviación*	How far a number is from the mean.	Mean 19, value 23 → 4
Absolute Value *Spanish: Valor absoluto*	How far a number is from zero. It is never negative.	Distance is never below zero
Spread *Spanish: Dispersión*	How far apart the numbers are.	Small MAD = tight set
Data distribution *Spanish: Distribución de datos*	How the data looks: where it sits and how spread out it is.	A set clustered tightly around the mean has low MAD; a set spread far from the mean has high MAD
Variability *Spanish: Variabilidad*	How spread out the numbers are.	Low variability (MAD = 1): very consistent. High variability (MAD = 8): very spread out

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Four plants grew 3, 7, 8 and 10 cm. Find the MAD.

1. Mean: 28 ÷ 4 = 7.
2. Distances from 7: 4, 0, 1, 3.
3. Sum = 8, and 8 ÷ 4 = 2.

Answer: MAD = 2 cm

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New data set: 9, 10, 11. First, what is the mean? Add the three numbers and divide by 3.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The mean of this set is _____ .
- The distance from _____ to the mean is _____ .
- The distances total _____ , and there are _____ values.
- The MAD is _____ , which means _____ .

Word bank mean absolute deviation | deviation | distance | absolute value | mean | spread | average | total | divide | typical | consistent | variation

WATCH OUT!

A common mistake

A common mistake in Mean Absolute Deviation is adding the signed deviations (value − mean) instead of their absolute values, so the positives and negatives cancel out to 0. For 6, 8, 10, 12 (mean = 9), the deviations are −3, −1, 1, 3

LESSON 2.9

Describe Data by Mean Absolute Deviation

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

The mean is 10. Place each data value on the number line, then find how far each is from the mean: Data: 7, 10, 13.



HINT 7 is 3 units below the mean. 10 is exactly at the mean. 13 is 3 units above the mean.

SHOW YOUR THINKING

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

The mean of a data set is 15. One value is 11. What is the absolute deviation of that value from the mean?

(A) 4
(B) −4
(C) 11
(D) 26

HINT Absolute deviation asks: how FAR is the value 11 from the mean of 15? A distance is never negative; zero is allowed.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A data set has absolute deviations of 3, 1, 5, 2, 4. What is the MAD?

(A) 1
(B) 3
(C) 5
(D) 15

HINT You're given the five absolute deviations already — no subtracting needed. MAD is their average.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The mean of a data set is 20. A value is 26. What is the absolute deviation?

(A) 6
(B) −6
(C) 26
(D) 46

HINT Same skill as before: absolute deviation is the distance between the value 26 and the mean 20.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Team A has MAD = 2.1 points. Team B has MAD = 6.8 points. Which team is more consistent?

(A) Team B — higher MAD means better performance

(B) Both are equally consistent

(C) Cannot tell from MAD alone

(D) Team A — lower MAD means scores are closer to the mean

HINT Consistent means scores stay close to the team's average game after game. MAD measures exactly that closeness.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Data set** A player's points: 6, 10, 8. Mean = (6 + 10 + 8) ÷ 3 = 8.
2. **Find each deviation (value − mean)** 6 − 8 = −2, 10 − 8 = 2, 8 − 8 = 0
3. **Find the MAD** MAD = (−2 + 2 + 0) ÷ 3 = 0 ÷ 3 = 0

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT The deviations −2 and 2 cancel to 0 — that's the warning sign. MAD is never 0 unless every value equals the mean.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Player A's MAD is 2.4 points. If Player B's MAD is 7.2 points, what does that tell you about their consistency?

SENTENCE STARTER Player A has a ___ MAD, which means their scores are ___ to the mean. Player B has a ___ MAD, which means their scores are ___ from the mean. The more consistent player is ___ because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.9

Describe Data by Mean Absolute Deviation

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Deviation Detective: the quiz scores are 12, 8, 10, 14, 6 and the mean is 10. Complete each deviation and absolute deviation, then compute the MAD.

Score	Deviation (Score − Mean)	Absolute Deviation
12		
8		
10		
14		
6		

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each MAD value to what it tells you about the data.

_____ MAD = 1.5

_____ MAD = 8.0

_____ MAD = 0

_____ MAD = 4.2

(A) Very consistent data

(B) Very spread out data

(C) All values are identical

(D) Moderate spread

3 **Write the name of the correct group on each line.**

Rank from least to most spread out.

Groups: Least to most spread out

_____ Set A: 9, 10, 10, 11 (MAD ≈ 0.5)

_____ Set B: 5, 10, 10, 15 (MAD = 2.5)

_____ Set C: 1, 10, 10, 19 (MAD = 4.5)

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Two runners have the same average time of 60 seconds. Runner A's MAD is 1.2 seconds. Runner B's MAD is 5.8 seconds. Which runner is the coach more likely to pick for a relay race that needs a reliable time?

(A) Runner B — higher MAD means faster potential

(B) Either — they have the same average

(C) Runner A — lower MAD means more consistent times

(D) Neither — MAD doesn't matter for relay races

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Data set** A swimmer's lap times: 2, 5, 8, 9. Mean = (2 + 5 + 8 + 9) ÷ 4 = 24 ÷ 4 = 6.
2. **Find each absolute deviation** |2 − 6| = 4, |5 − 6| = 1, |8 − 6| = 2, |9 − 6| = 3
3. **Find the MAD** Sum of absolute deviations = 4 + 1 + 2 + 3 = 10. MAD = 10 ÷ 3 ≈ 3.33

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Player A's MAD is 2.4 points. If Player B's MAD is 7.2 points, what does that tell you about their consistency?

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Challenge · Extension

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each scenario by whether low or high variability is desirable.

Groups: Want LOW MAD (consistent) | Want HIGH MAD (lots of spread)

_____ A factory making bolts that must be exactly 5 cm long

_____ A bus schedule where arrival times should be reliable

_____ A game show where contestants try to score as many different point values as possible

PART 2 Explain and correct

reasoning

2 **Find the mistake. Explain it, then show the correct work.**

Find the MAD Error

1. **Data set** 5, 10, 15, 20, 25
2. **Find mean** Sum = 75, Count = 5, Mean = 15
3. **Find deviations** −10, −5, 0, 5, 10
4. **Find MAD** MAD = (−10 + −5 + 0 + 5 + 10) ÷ 5 = 0

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 **Answer in complete sentences.**

Two basketball teams both average 75 points per game. Team A's last 5 scores: 73, 76, 74, 77, 75. Team B's last 5 scores: 60, 90, 65, 85, 75. Calculate the MAD for each team and explain which team a coach would prefer if they need predictable scoring.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Player A's MAD is 2.4 points. If Player B's MAD is 7.2 points, what does that tell you about their consistency?

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